SDUHSD CAREER TECHNICAL EDUCATION

Machine Learning

Campus: Canyon Crest Academy
CTE Industry Sector: Information and Communication Technologies
CTE Pathway: Software and Systems Development

Course Sequence: Enrichment
Prerequisites: AP Computer Science A OR AP Computer Science Principles
10 credits, UC Approved Area D
Recommended Pathway Sequence:

Course 1	Course 2	Course 3	Enrichment
Computer Science Foundations OR Computer Game Design	Human-Computer Interaction (HCI) Design		AP Computer Science Principles AP Computer Science A Machine Learning

Standards:

- CA Computer Science Standards: https://www.cde.ca.gov/be/st/ss/computerscicontentstds.asp
- CTE Computer Science curriculum model standards: https://www.cde.ca.gov/ci/ct/sf/documents/infocomtech.pdf

Course Description:
Informatics, or "data science," and the use of Artificial Intelligence technologies is growing rapidly in our society. Just a few AI applications include self-driving cars, personal assistants, product recommendations, robotics, data analysis, and web searching. These applications involve self-learning systems that are trained based on large amounts of data which usually includes intelligence based on algorithms.

This semester course provides students exposure to the field of informatics, via problem-solving applied to the varied fields of Informatics such as Artificial Intelligence, Data Mining/Analysis, and Social Computing. Uniting application-specific approaches in Cognitive Systems, Machine Learning, and Robotics, students explore creative applications of algorithms to a wide range of real-world challenges in math, science, language, and art. Projects in this course concentrate on developing interdisciplinary applications of

information science, especially in its application of Artificial Intelligence with a focus on the design and implementation of intelligent agents that perform tasks with some degree of autonomy.

- Students will acquire the fundamentals of programming in the Python Programming Language to facilitate developing applications capable of retrieving, searching, manipulating, analyzing, and displaying data
- Students gain competency and fluency with the vocabulary of algorithms, informatics, big data, and artificial intelligence through exposure to a variety of sources and diverse perspectives.
- Students will develop critical thinking, problem solving, computational thinking, effective communication, and teamwork skills. Students will express their understanding of the legal, policy, and ethical factors through class discussion of a weekly topic, conducting research on a related topic, and a presentation of their research to the class.

Course Syllabi and Standards Map:

Unit	Name	Topics	CA CTE Model Curriculum Standards	Projects
1	Review of Programming concepts with Python	Fundamentals of programming, integrated development environment (IDE), and visualization	C2.0: Define and analyze systems and software requirements. C2.5: Determine and document the requirements and alternative solutions to fulfill the customers' needs. C4.0 Develop software using programming languages C5.0 Test, debug, and improve software development work.	Create an Age guessing game, Graphical racetrack, Google Colab IDE

			C9.4 Apply the concepts of embedded programming, including digital logic, machine-level representation of data, and memory-system organization.	
		Data structures and random numbers		Implement a Pseudorandom number generator

		Object Oriented Programming (OOP)		Create a deck of cards and apply to game Blackjack using OOP techniques
				
				● The Expected Learning Outcomes (ELO's) of this unit include the following: Students will demonstrate competency in the ELOs by writing Python-based programs, including but not limited to the following as mapped to the topics in this unit (see unit lab activities for example project details):
				● The following lab activity describes the OOP topic in this unit.
				  ● Inquiry: Prompt students with the following question: How would you create a program that could play the card game blackjack
				  ● Observation: Students research the basic rules of blackjack, then play the game with a physical deck of cards. Students research conventional rules on when a player and dealer should “hit” (take a

				card) and “hold” (stop taking cards) and write down the rules. Students create an alternative set of dealer rules and write up the behavior rules (e.g. Programming conditional statements) for how a dealer reacts to the cards in hand. ● Implementation (Connection to Computer Science): Using the Python programming language in the Google Colab or Jupiter Notebook integrated development environment, students create the following: a card class with attributes suit, value, rank. a deck class comprised of a collection of cards using a Python collections data structure (e.g. Python language list or dictionary). A shuffle method to rearrange the card order according to a random number generator. Students write code implementing the control logic reflecting the conventional and alternative rules for the dealer to “hit” and “hold” based on the Observation rules. Students write the control logic code to input a user’s choices for hitting and holding, and the method(s) to determine if the dealer or player wins the hand. ● Analysis: Students run the program and confirm the control logic determined in the Observation step and written in the Implementation step. Students record statistics on how many times the player vs. dealer wins a hand across several deck initializations. ● Write-up: Students write a reflection of how the implemented dealer rules affected the win/loss statistics for both the conventional and alternative rules. The

				different rule sets are compared and contrasted.
2	Algorithms	Types of algorithms, ethics, research presentations	C4.10 Create and know the comparative advantages of various queue, sorting, and searching algorithms.	• Research and present at least 2 different types of algorithms • Research and present ethical concerns with different algorithms (e.g. deep fake technology, and web scraping) • Research and find at least 2 AI based computer games and present the algorithms used in its implementation
		Web Scraping		Web Scraping Project: • Students access and collect website information about the company and products. In this project, students will create a 'web scraper' to access and retrieve real-time data from the US Geological Service (USGS) reflecting the latest active earthquakes around the world which are equal or above a user input magnitude. • Inquiry: Prompt students with the following question: How many earthquakes each day occur on earth, and which regions have the most over a period of time. • Observation: Students visit and become familiar with data reported from the USGS site https://earthquake.usgs.gov/earthquakes/feed/ Students research the 'json' format as it relates to data transmission over computer science applications originating in the cloud. Students research the Python machine learning libraries 'Numpy, and 'Pandas' for information on reading and analyzing datasets. • Implementation (Connection to Computer Science): Using the Python programming language in the Google Colab or Jupiter Notebook integrated development environment, students setup program imports and prompt the user for the magnitude from

				which the USGS data will be accessed. That is, any earthquake greater than or equal to the input magnitude will be retrieved from the database. Student create Python methods to access, convert the data from json to formatted output, then print the output of the data you retrieved from the site: http://earthquake.usgs.gov/earthquakes/feed/v1.0/summary/2.5_day.geojson. Students then convert the output to CSV format (for spreadsheet analysis), and read the data into the Machine learning library 'Pandas' format for statistical analysis. ● Analysis: Students observe the outputs and provide intuitive analysis on where earthquakes occur most. Students then evaluate the data in a spreadsheet and find conventional statistics (mean, standard deviation, min, max) for the earthquakes over a given time interval, then confirm using the Python library Pandas functions. ● Write-up: Students reflect upon the data analyzed and write a report showing the earthquake "hot spots" in the word over a given time interval for both frequency, and magnitude.
3	Informatics	Image Classification	C8.0 Develop databases. C9.0 Develop software for a variety of devices C10.0 Develop intelligent computing.	● Use image classifier model and build into a javascript application ● Implement object detector using Coco MobileNet and Raspberry Pi

		Statistics		● Hypothesis testing
				  ● Use Python to Calculate Z-score from experimental data
				  ● Use Python to Perform Chi-squared tests from experimental data
				● Introduction to Numpy and Pandas libraries for use with Statistics problems

		Regression		• Perform Simple and Multiple Linear regression on actual unemployment data from government reports
				• Perform polynomial regression on weather prediction (temperature and rainfall) based on released government reports
				  • Inquiry: Prompt students with the following question: How is Atmospheric pressure (independent variable) related to Humidity (dependent variable) in real world data obtained by the US weather service.
				  • Observation: Students visit the national weather service https://www.weather.gov/ , and find downloadable datasets on the site. Student examine datasets and general information provided on the site related to weather patterns and parameters collected (e.g. humidity, temperature, atmospheric pressure). Students research the Python machine learning libraries 'Numpy, and 'Pandas' for information on reading and analyzing datasets.
				  • Implementation (Connection to Computer Science): Using the Python programming language in the Google Colab or Jupiter Notebook integrated development environment, students write methods to read in the data feature independent variables (X), and the dependent variable Humidity (y) from the Pandas 'weather' table, create a function that generates a polynomial regression model for the input dataset and generated a 'fit' function to fit the regression function to the data.
				  • Analysis: Students call the methods written in the Implementation step from a main Python program for the degrees

				1,2,3,4 and print out results (coeffs, intercept, r2 score). ● Write-up: Students write a report reflecting on how the regression functions performed for various degrees. ● Extension: Students run the methods created in the Implementation step for 2 other variables chose from the weather dataset and re-analyze the results.
4	Artificial Intelligence Ethics	History and foundations	C3.0 Create effective interfaces between humans and technology. C10.3 Know artificial intelligence methods such as neural networks, Bayesian inferences, fuzzy logic, and finite state machines.	● Research and implement a Machine Learning "fairness" algorithm ● Research and implement a Bias algorithm ● Research and implement Model Cards on a Machine Learning algorithm
		Machine Learning Algorithms	C10.4 Implement artificial intelligent behavior through various methods: mathematical modeling, reinforcement learning, and probabilistic analysis.	● Implement supervised classification algorithm (KNN) in Python ● Implement supervised classification algorithm (K-means) in Python ● Implement decision trees in Python: Analysis of deaths on the Titanic ● Inquiry: Prompt students with the following question: What were your odds of dying in the cruise ship Titanic disaster based on your gender, age, or social class status? ● Observation: Students research the Titanic sinking on May 31, 1911. Students are provided datasets to train and test machine learning algorithms which have been covered in previous sub-units. Students analyze the data by hand to try to identify trends or predictions and write up their predictions.

				• Implementation:(Connection to Computer Science): Using the Python programming language in the Google Colab or Jupiter Notebook integrated development environment, students write methods to read in the training and test datasets. Students then write methods to implement the following functions: constructors to create decision tree nodes, calculate entropy from class labels, calculate information gain from decision tree split points, recursively grow the decision tree (i.e. the fit function), predict outputs with test data after training. • Analysis: Students call the methods written in the Implementation step from a main Python program and compare the results with what they had predicted in the Observation step. • Write-up: Students write a report reflecting the results of the Analysis step and how it differed from their predictions in the Observation step. Students reflect on the use of the decision tree algorithm to analyze the chances of dying on the Titanic. • Extension: Students use Python machine learning visualization functions to create a graphical representation of the decision tree created in the Implementation step.
		Semester project		• Students research a Machine Learning model and present to the class the description of the app and the machine learning algorithm(s) used. Students research the algorithm and provide implementation details.

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Grading: A variety of assessments will be used including journal entries, projects and self and peer assessments. For example students will be assessed on their journal entries which include warm-up exercises to the 3dB point on a filter. Another example assessment includes student demonstration of a project implementing the combinatorial logic associated with a fire alarm. Another example assessment includes a student self-evaluation of the design of his or her sequential timer. Another example assessment includes peer review and voting of students' final class project.

Work load: (High, Medium, Low)

Class setting: This relates directly to how much the student is expected to work independently or in a group requiring self-initiated work, problem solving, etc versus and highly guided and structured class).
(Highly independent, Moderately Independent, Structured). Hands-on projects accompany each unit.
*Each year all CTE courses are reviewed by industry professionals within the pathway to ensure the most current, relevant experience for our students